

Delta Lake MERGE Implementation using Pandas and PySpark

Assignment Report

Field	Details
Name	Eklavya
Course	B.Tech Computer Science & Engineering
Organization	Celebal Technologies
Assignment	Delta Lake MERGE Implementation
Platform	Databricks Free Edition
Technologies	Python, Pandas, PySpark, Delta Lake

Introduction

Data preprocessing is an important step before analysis and data engineering tasks. This assignment demonstrates data cleaning using Pandas followed by incremental data processing using Delta Lake MERGE in Databricks.

Objectives

- Load the CSV dataset.
- Explore the dataset.
- Handle missing values.
- Remove duplicate records.
- Create derived columns (Price and total_amount).
- Save the cleaned dataset.
- Load the cleaned data into Delta Lake.
- Create master and incremental tables.
- Perform Delta Lake MERGE.
- Validate the final results.

Dataset

The Sample Superstore dataset from Kaggle was used. It contains order, customer, product, sales, quantity, discount and profit information with 9,994 records.

Part A – Pandas Data Cleaning

The dataset was loaded into a Pandas DataFrame and explored using `head()`, `shape`, `columns` and `data types`. Missing values were checked, duplicate records were removed, filtering and column selection were demonstrated, and two derived columns

(Price and total_amount) were created. The cleaned dataset was exported as cleaned_superstore.csv.

Part B – Delta Lake MERGE

The cleaned dataset was loaded into Spark, column names were renamed for Delta compatibility, and a Delta table named customer_master was created. An incremental dataset (customer_incremental) containing updated and new records was generated. The Delta MERGE operation updated existing records and inserted new records into the master table.

Validation

Original Records: 9994

Incremental Records: 10

Final Records after MERGE: 9999

Duplicate Row_ID Records: 0

The final dataset was verified through record count, duplicate validation, updated records and newly inserted records.

Challenges Faced

- CSV parsing with quoted values.
- Delta Lake column naming restrictions.
- Data type validation.
- Databricks Serverless compatibility.

These issues were resolved by correctly reading the CSV, renaming columns, validating the schema and recreating the Delta table.

Conclusion

The assignment successfully demonstrated data cleaning with Pandas and incremental data processing using Delta Lake MERGE. Existing records were updated, new records were inserted, and the final dataset was validated successfully.

References

1. Kaggle – Sample Superstore Dataset
2. Microsoft Learn – Delta Lake MERGE Documentation